```
  def each
    @string.scan(/[aeiou]/) do |vowel|
      yield vowel
    end
  end
end
```

tut_modules/vowel_finder_eg.rb

```
require_relative "vowel_finder"
vf = VowelFinder.new("the quick brown fox jumped")
puts vf.reduce(:+)
```

produces:

```
euiooue
```

Note we used the same pattern in the call to reduce in these examples—we're using it to perform a summation. When applied to numbers, it returns the arithmetic sum; when applied to strings, it concatenates them. We can use a module to encapsulate this functionality too:

tut_modules/vowel_finder_sum.rb

```
require_relative "vowel_finder"

module Summable
  def sum
    reduce(:+)
  end
end

class Array
  include Summable
end

class Range
  include Summable
end

class VowelFinder
  include Summable
end

puts [1, 2, 3, 4, 5].sum
puts ("a".."m").sum

vf = VowelFinder.new("the quick brown fox jumped")
puts vf.sum
```

produces:

```
15
abcdefghijklm
euiooue
```

Note that you don't need to define this particular example in Ruby since sum is already defined as part of the Enumerable module.

Instance Variables in Mixins

People learning Ruby often ask, "What happens to instance variables in a module that is used as a mixin?